

can be achieved by correcting the perturbed dose using TLD correction factors calculated by Monte Carlo (MC) methods. The current study investigates the MC calculated TLD correction factors against depth and field size within a proposed lung phantom.

Materials and Methods: 14 TLD-100 measurements were done along the central axis within a lung phantom, which consists of CIRS plastic water slabs sandwiching 14 pieces of composite cork slabs. TLD correction for the positions in composite cork was determined by taking the ratio of the Monte Carlo simulated dose response of TLD in cork to the Monte Carlo simulated dose response of TLD in water. BEAMnrc and DOSXYZnrc user codes were used for simulation (EGSnrc, National Research Council of Canada, Ottawa, ON).

Results: It was found that the greatest TLD correction was for the smallest field size investigated, $2 \times 2 \text{ cm}^2$, where electronic disequilibrium was found to be the greatest as shown in Figure 1. Corrections at inhomogeneous interfaces were dependent on the medium before the interface. In addition, MC corrected TLD measurements agree with Acuros External Beam (AXB) $2 \times 2 \text{ cm}^2$ predicted depth doses to within 2% as shown in Figure 2.

Discussion: TLD perturbation is dependent on three main aspects, namely, lateral charge particle disequilibrium, over response of TLD in broad beam and partial volume averaging. Due to the small physical size of TLD-100, partial volume averaging was deemed negligible in this study. Lateral charge particle disequilibrium was found to be dependent on field size, the TLD material, and the density of the phantom. The electron range in low-density materials, such as composite cork, is longer than the range in water. Hence, lateral electronic disequilibrium occurs predominantly in composite cork. Furthermore, under small field conditions, the Compton electronic range is greater than the distance between the point of measurement to the field edge, increasing the probability for an electron to transfer its energy outside the field, resulting in greater electronic disequilibrium.

Lastly, the over response of TLD in broad beam was due to TLDs having a higher atomic number than that of water. 6 MV photon beam consists of an extensive amount of low-energy photons as compared to higher energies photon beam. Due to larger cross-section for photoelectric effect in high atomic

AB-3

TLD CORRECTION FACTORS FOR FIELD SIZES USED IN LUNG STEREOTACTIC BODY RADIATION THERAPY DOSIMETRIC MEASUREMENTS

Roger Cai Xiang Soh^{1,2}, Guan Heng Tay³,
Wen Siang Lew¹, Shaun Baggarley²,
James Cheow Lei Lee^{1,3}

¹Division of Physics and Applied Physics, School of Physical and Mathematical Sciences, Nanyang Technological University, ²Department of Radiation Oncology, National University Cancer Institute Singapore, National University Hospital, ³Division of Radiation Oncology, National Cancer Centre Singapore, Singapore. E-mail: roger_cai_xiang_soh@nuhs.edu.sg

Introduction: TLD dosimetric verification for Lung Stereotactic Body Radiation Therapy (SBRT) may be a challenge due to small field TLD dose perturbation within the low-density lung medium. Accurate measurement

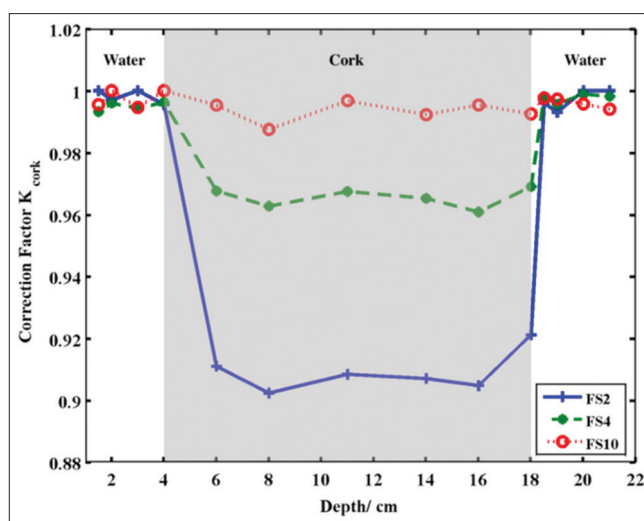


Figure 1: Monte Carlo simulated TLD correction factors with respect to depth

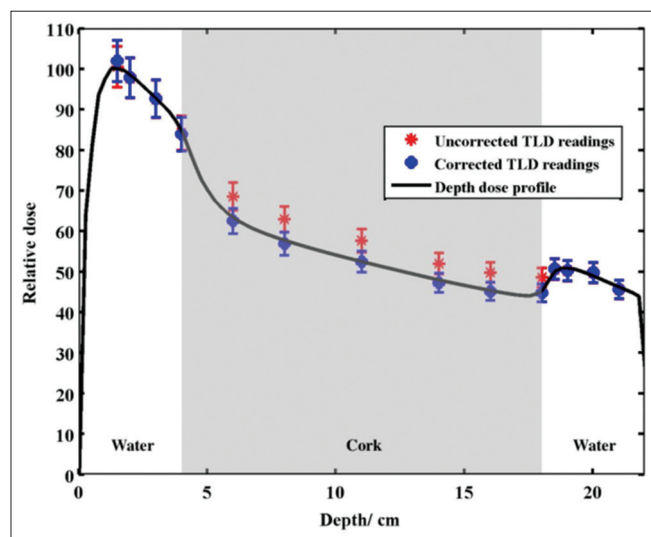


Figure 2: Monte Carlo TLD corrected measurements against AXB 2 x 2 cm² predicted dose profile

number materials within TLDs, low-energy photons will be absorbed to a high extent. This low energy photon absorption increases with increasing field sizes.

Conclusion: The results presented in this study indicate that corrections will be required for in-vivo TLD measurements in the proposed lung phantom for Lung SBRT dosimetry measurements. Monte Carlo simulated TLD correction factors calculated were dependent on the depth of measurement and field sizes. The greatest TLD correction was for the smallest field size investigated, 2 x 2cm². Further studies may include the effect of TLD angular dependence on TLD correction factors.